# Task-Aligned Equivariant Gates for Variational Quantum Learning

Author One, Author Two, and Author Three
LMU Munich
Corresponding author: TODO

***Abstract*—Symmetry is one of the cleanest ways to inject inductive bias into variational quantum learning models: if the task is invariant, the circuit should not spend data learning that invariance. Meyer et al. showed this principle on Tic-Tac-Toe by enforcing the full dihedral symmetry $D_4$ of the board. We ask what comes next once full equivariance is available. Is the relevant design variable only whether the full symmetry is imposed, or can equivariance itself be shaped into a more task-aligned architecture? We study this question on the same nine-qubit Tic-Tac-Toe benchmark, using data re-uploading circuits with subgroup-based parameter sharing, matched random-sharing controls, and additional equivariant gates acting on the eight winning lines. The results reproduce the expected Meyer-style behavior: full $D_4$ improves test accuracy over no symmetry and reduces the generalization gap. Partial symmetry is not dominant, but $C_4$ closely tracks $D_4$ across data regimes. The strongest effect comes from keeping equivariance while adding winning-line interactions: the best $D_4$-equivariant edge-plus-line circuit reaches about $0.80$ test accuracy, compared with about $0.69$ for the Meyer-style edge circuit. These results suggest that equivariance should be treated not merely as parameter reduction, but as an architecture design principle for placing trainable interactions where the task structure lives.**

***Index Terms*—variational quantum learning, equivariance, symmetry, inductive bias, quantum machine learning**

## I. Introduction

A variational quantum learning model is defined as much by what it is not allowed to learn as by what it can express. Modern variational quantum algorithms expose a large space of trainable circuits [1], and data re-uploading makes those circuits flexible enough to act as nonlinear classifiers [2], [3]. If a classification task is invariant under rotations, reflections, or permutations, however, a generic circuit can waste scarce training data fitting distinctions that the label will never use. Symmetry-aware circuits remove those distinctions by construction. This is the central lesson of Meyer et al. [4]: for Tic-Tac-Toe, whose winner is unchanged by the eight rotations and reflections of the square, a $D_4$-equivariant variational quantum learning model generalizes better than a model built from unconstrained gates.

That result leaves a second design question open. Once the full task symmetry is known, should it simply be imposed as strongly as possible, or should symmetry be treated as a tunable architectural resource? In the language of geometric learning [5], equivariance restricts how representations may change under transformations. Full equivariance reduces the hypothesis class, but it does not decide which equivariant interactions should be available. Conversely, a subgroup such as $C_4$ preserves rotation structure while relaxing reflection constraints. These options form a family of inductive biases between the unconstrained model and the fully equivariant one. The practical question is therefore not only "symmetry or no symmetry," but where useful equivariant capacity should be placed.

We study this question in the same controlled setting as Meyer et al.: balanced Tic-Tac-Toe board classification into circle win, draw, and cross win. The task is deliberately small and classically trivial. Its purpose is not to claim quantum advantage, but to provide a finite, exactly understood benchmark where the symmetry group is known and every architectural change can be audited. We keep the Meyer-style data encoding, invariant observables, and directed edge interactions as the reproduction anchor, then extend the ansatz in two directions.

First, we replace the binary comparison between no symmetry and full $D_4$ with subgroup-equivariant parameter sharing. We compare no sharing, two $Z_2$ subgroups, the rotation group $C_4$, the order-four subgroup $D_2$, and full $D_4$. This tests whether less symmetry can preserve useful expressivity while retaining geometric bias. Second, we add task-aligned gates on the eight winning-line triples. These gates are still tied over group orbits, so they preserve equivariance, but they give the circuit direct access to the three-cell structure that determines Tic-Tac-Toe wins.

Our contribution is empirical and architectural. We reproduce the qualitative Meyer-style result that $D_4$ improves generalization over an unconstrained baseline. We then show that $C_4$ nearly matches $D_4$ in the original edge ansatz, while parameter-matched random sharing performs substantially worse than symmetry-aware sharing. Finally, we show that a $D_4$-equivariant edge-plus-winning-line ansatz improves test accuracy from roughly $0.69$ to roughly $0.80$ in the same training protocol. The lesson is not that the full symmetry is wrong. It is that symmetry is only the outer constraint; within it, task-aligned equivariant gates can matter more than the choice between full and partial sharing.

## II. Protocol

The board is indexed as

$$\begin{matrix} 0 & 1 & 2 \\ 7 & 8 & 3, \\ 6 & 5 & 4 \end{matrix}$$

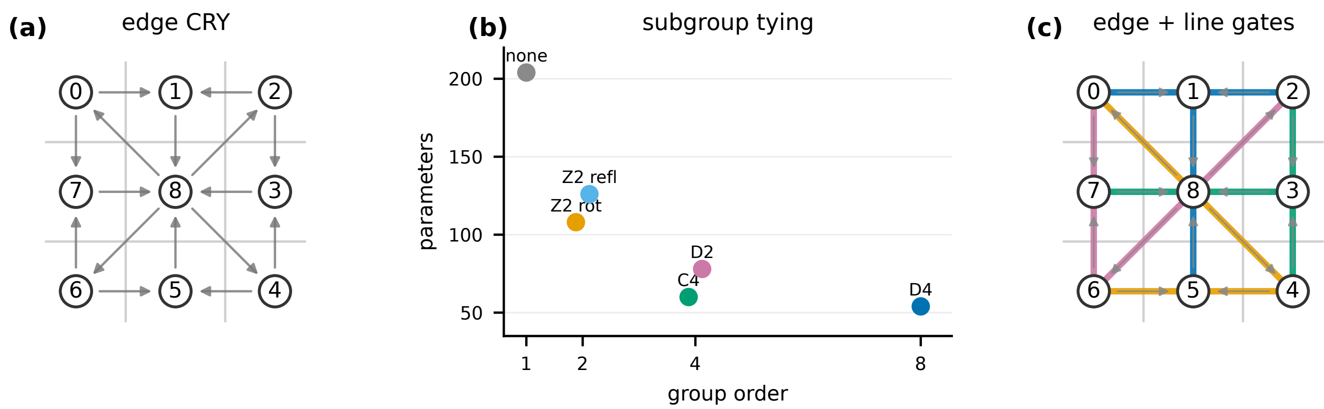


Fig. 1. Architecture design space. (a) The Meyer-style baseline uses directed CRY interactions on board edges and center connections. (b) Subgroups of $D_4$ induce different parameter-sharing orbits, changing the number of trainable parameters without changing the board geometry. (c) Our oracle-inspired extension keeps equivariance but adds trainable interactions on the eight winning-line triples.

so that corners are $(0, 2, 4, 6)$, edges are $(1, 3, 5, 7)$, and the center is $8$. Each legal reachable board state is encoded as $x_i \in \{-1, 0, +1\}$, with $+1$ for cross, $-1$ for circle, and $0$ for empty. Terminal circle wins, draws, and cross wins are labeled by vectors $(+1, -1, -1)$, $(-1, +1, -1)$, and $(-1, -1, +1)$, respectively. Following Meyer et al., unfinished games are included and labeled as draws, and train/test sets are balanced across the three classes.

Each model uses nine qubits initialized in $|0\rangle^{\otimes 9}$. The data embedding applies

$$U(x) = \prod_{i=0}^{8} R_X(2\pi x_i/3)$$

and is re-uploaded in every layer. A circuit has $L = 3$ re-uploading layers and $p = 2$ trainable repetitions per layer unless stated otherwise. The single-qubit block is the paper-style $R_X R_Y$ block used throughout our main experiments, and the original edge ansatz applies directed CRY gates on the sixteen board pairs shown in fig. 1. Predictions are the invariant observable vector

$$\hat{y}(x) = \left( \frac{1}{4} \sum_{i \in C} \langle Z_i \rangle,\ \langle Z_8 \rangle,\ \frac{1}{4} \sum_{i \in E} \langle Z_i \rangle \right),$$

where $C$ are corners and $E$ are edges. The predicted class is the largest component of $\hat{y}$.

For a subgroup $H \leq D_4$, trainable parameters are shared over orbits of qubits and directed gate pairs under $H$. Thus the circuit topology is held fixed while the tying pattern changes. The six sharing patterns are no symmetry, $Z_2$ generated by a 180° rotation, $Z_2$ generated by a vertical reflection, $C_4$ rotations, $D_2$ generated by 180° rotation and horizontal/vertical reflections, and full $D_4$. With the $R_X R_Y + \text{CRY}$ block, the corresponding parameter counts for $L = 3, p = 2$ are 204, 108, 126, 60, 78, and 54.

The oracle-inspired families add trainable interactions on the eight winning triples: three rows, three columns, and two diagonals. We test MultiRZ-style $ZZZ$ rotations, symmetric controlled-controlled $R_Z$ interactions, and their combination, either alone or in addition to the original edge CRY layer. Parameters are again tied over subgroup orbits, so the resulting models remain equivariant under the selected subgroup.

Tab. I
PROTOCOL AT A GLANCE.

| | |
|---|---|
| Task | Tic-Tac-Toe winner classification |
| Encoding | $R_X(2\pi x_i/3)$ on 9 qubits |
| Readout | corner, center, edge $Z$ observables |
| Loss | mean squared error on label vectors |
| Optimizer | Adam, learning rate 0.01 |
| Default depth | $L = 3$, $p = 2$ |
| Batching | batch size 15, 30 steps/epoch |
| Training | 100 epochs, 5 seeds in draft runs |
| Simulator | PennyLane `lightning.qubit` |

## III. Evidence

Figure 2 summarizes the main evidence. The reproduction anchor behaves as expected. At training size 450 and test size 600, the unconstrained edge ansatz reaches mean test accuracy $0.614 \pm 0.029$, while the $D_4$-equivariant edge ansatz reaches $0.688 \pm 0.036$ over five seeds. The generalization gap drops from 0.094 to 0.023. The unconstrained circuit has more parameters, but the additional freedom mostly appears as overfitting rather than useful test performance.

The subgroup sweep shows a more nuanced picture. Full $D_4$ is not beaten robustly by a partial group, so the conservative conclusion is not that less symmetry is better. Instead, $C_4$ tracks $D_4$ closely across data regimes: at 30 training examples, $C_4$ obtains 0.552 test accuracy versus 0.531 for $D_4$; at 120 examples, the two are essentially tied at 0.630 and 0.629; at 450 examples, $C_4$ and $D_4$ again match at 0.689 and 0.688. We interpret this as evidence that the useful inductive bias is not a single binary setting. Rotation-only equivariance can preserve nearly the same generalization behavior while relaxing part of the full constraint.

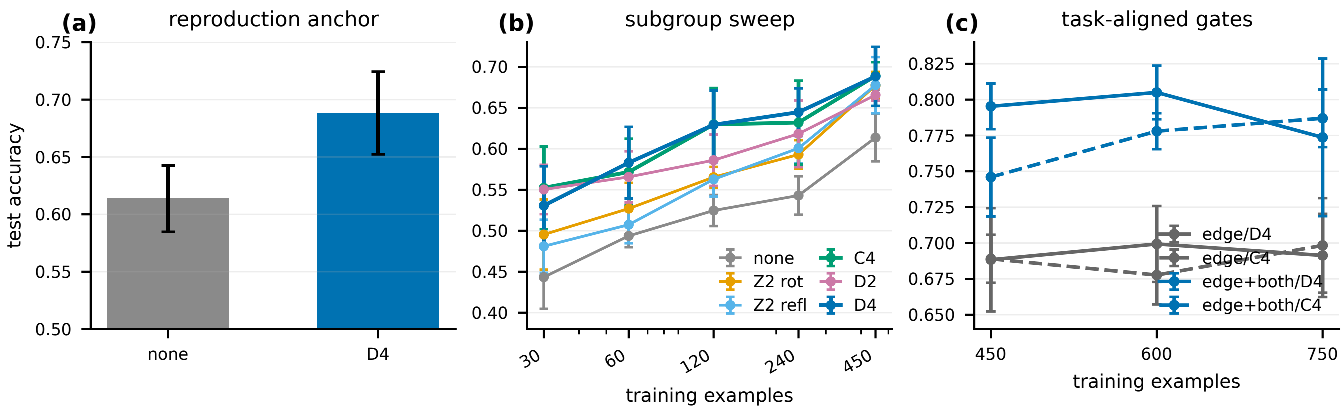


Fig. 2. Main empirical story. (a) Full $D_4$ equivariance improves test accuracy over no symmetry in the Meyer-style edge ansatz. (b) Across data regimes, $C_4$ closely follows $D_4$, while weaker or unconstrained sharing tends to lag. (c) The largest gain comes from keeping equivariance and adding task-aligned winning-line interactions. Error bars are 95% confidence intervals over five seeds.

Tab. II
KEY TEST-ACCURACY RESULTS USED IN THE CLAIMS. VALUES ARE MEANS WITH 95% CONFIDENCE INTERVALS OVER FIVE SEEDS.

| Model | Train | Params | Test acc. |
|---|---|---|---|
| edge/none | 450 | 204 | $0.614 \pm 0.029$ |
| edge/$D_4$ | 450 | 54 | $0.688 \pm 0.036$ |
| edge/$C_4$ | 450 | 60 | $0.689 \pm 0.017$ |
| random/$D_4$ | 450 | 54 | $0.572 \pm 0.026$ |
| edge+both/$D_4$ | 450 | 90 | $0.795 \pm 0.016$ |
| edge+both/$D_4$ | 600 | 90 | $0.805 \pm 0.019$ |

The strongest result is the oracle-inspired architecture. The Meyer-style edge/D4 model is around $0.69$ test accuracy in the robust sweep. Adding both winning-line interaction channels to the edge circuit increases performance to $0.795 \pm 0.016$ at 450 training examples and $0.805 \pm 0.019$ at 600 examples. The C4 version of the same architecture is also strong, reaching $0.778 \pm 0.013$ at 600 examples and $0.787 \pm 0.020$ at 750 examples. The improvement is therefore not obtained by breaking equivariance; it is obtained by using equivariance to tie a more task-aligned set of gates.

Figure 3 separates this effect from two simpler explanations. First, symmetry-aware sharing is not equivalent to arbitrary parameter reduction. At matched parameter counts, random tying is consistently worse: for $D_4$, symmetry-aware sharing gives $0.688$ while random sharing gives $0.572$; for $C_4$, the corresponding numbers are $0.689$ and $0.551$. Second, the oracle-inspired gain is not simply "more parameters." Several models with similar or fewer parameters improve over the edge baseline, but the best result appears when the original edge interactions are combined with both winning-line channels. The parameter-count scatter shows the same point visually: the best circuit is not just farther to the right, it moves into a different accuracy region.

## IV. DISCUSSION

The results support a more architectural view of equivariance in variational quantum learning. Meyer et al. showed that enforcing the full task symmetry can improve generalization. Our reproduction confirms that behavior in the same Tic-Tac-Toe setting. But once the model is already equivariant, the remaining design space is still large: one must decide which equivariant interactions are available. The winning-line experiment shows that this choice can matter more than the difference between no symmetry and full symmetry in the original edge ansatz.

This also clarifies what the partial-symmetry experiments do and do not show. They do not establish that $C_4$ is generally better than $D_4$. With five seeds, the honest statement is that $C_4$ is competitive and sometimes slightly ahead, while $D_4$ remains the more conservative default. The value of the partial sweep is that it turns symmetry into a tunable axis and shows that nearby subgroup biases can have similar generalization behavior despite different parameter counts and constraints.

The random-sharing control is essential for the claim. If symmetry-aware tying and random tying performed similarly, then the story would reduce to ordinary regularization by parameter count. They do not. Group-orbit sharing retains substantially more test accuracy than random sharing at the same number of parameters. This supports the interpretation that the circuit benefits from the structure of the board symmetry, not merely from being smaller.

The oracle-inspired ansatz should also be read conservatively. Tic-Tac-Toe is classically solved by a deterministic rule, and our separate rule oracle indeed has zero trainable parameters and perfect accuracy. We do not claim quantum advantage, hardware readiness, or a universal recipe for all symmetric tasks. The claim is narrower and, for that reason, sharper: for a fixed symmetric benchmark and a Meyer-style variational quantum learning setup, task-aligned equivariant gates can substantially improve generalization while preserving the symmetry principle.

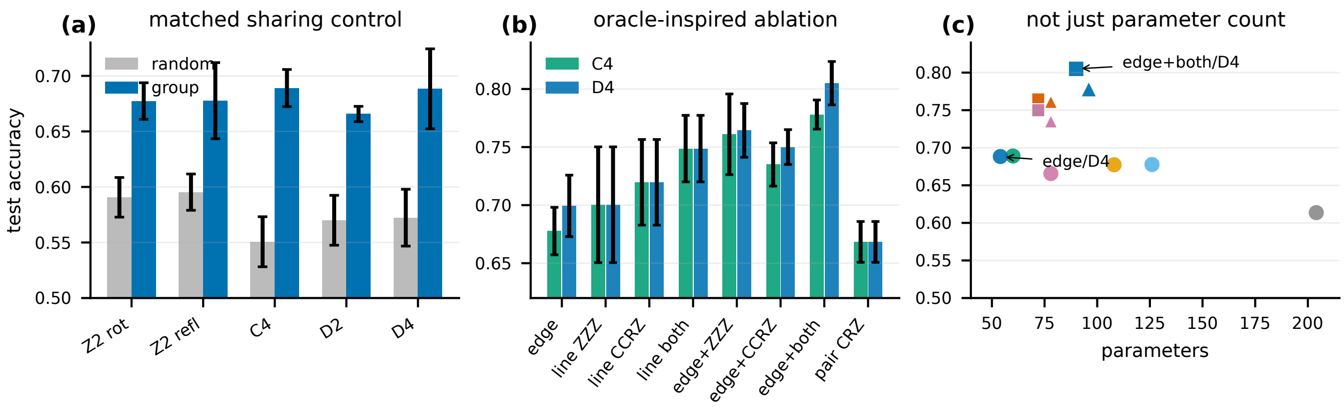


Fig. 3. Controls and ablations. (a) Parameter-matched random sharing underperforms group-orbit sharing, so the benefit is not only fewer parameters. (b) Oracle-inspired ablations at 600 training examples show that the strongest model combines edge interactions with both winning-line channels. (c) Accuracy is not explained by parameter count alone; task-aligned equivariant geometry is the relevant change.

The useful analogy is not to a larger neural network that wins by brute force. The edge-plus-line model changes the basis of interactions. The original edge ansatz sees local adjacencies around the board and connections to the center; the winning-line ansatz sees the three-cell motifs from which the label is actually computed. Both are equivariant under the same board group, but they expose different invariant features to the readout. In this sense the result is closer to choosing a convolutional kernel that matches the object geometry than to increasing width.

This distinction matters for future variational quantum learning benchmarks. If a symmetric task has known local motifs, conserved substructures, or relational patterns, then a faithful equivariant circuit should represent those objects directly instead of only tying generic one- and two-qubit gates. The Tic-Tac-Toe case is useful precisely because the correct motifs are obvious: the eight winning lines. That makes the benchmark a controlled demonstration of a broader recipe rather than a hard application.

This makes the result a natural extension of Meyer et al. Their paper asks how to exploit a known symmetry in a variational quantum model. Our answer is that exploiting the symmetry is only the first step. The next step is to design the equivariant gates around the combinatorial structure of the task. In Tic-Tac-Toe, that structure is the set of winning triples. In other domains, it may be local motifs, conserved interaction patterns, or invariant substructures. The broader lesson is that equivariance is not merely a constraint imposed on a generic ansatz; it is a design language for building the ansatz itself.

## Reproducibility Appendix

The code and experiment artifacts are in the project repository https://github.com/eybmits/qc_symmetry. All numerical claims in this draft are computed from checked CSV files in `results/csv/`. The exact sources are the reproduction file `results_reproduction.csv`, the subgroup file `results_partial_data_sweep_full_L3p2.csv`, the random-sharing file `results_random_sharing_control_full_L3p2_train450.csv`, and the oracle-gate file `results_oracle_inspired_robust_C4D4_L3p2_train450_600_750.csv`. Figures are regenerated by running `python paper/make_figures.py`. The simulations use PennyLane [6], PyTorch [7], NumPy, Pandas, and Matplotlib. Unless otherwise stated, models use $L = 3$, $p = 2$, batch size 15, 100 epochs, 30 minibatch steps per epoch, Adam [8] with learning rate 0.01, and five seeds. The present draft is therefore a finite-seed simulation study; final submission should rerun the headline comparisons with 10–20 seeds.

The repository also contains experiment families that are not used as headline figures. The depth sweep tests whether the subgroup ordering changes with effective circuit depth $Lp$; the compression sweep ties parameters more aggressively than $D_4$ to check whether still fewer parameters are sufficient; and the approximate-symmetry runs deliberately corrupt labels on a fraction of $D_4$ orbits. These runs are useful for paper development, but the current evidence is less central than the three claims reported here: reproduction, structured sharing, and winning-line architecture.

Several sanity checks are implemented directly in the codebase. The data generator enumerates legal reachable Tic-Tac-Toe states and verifies that labels are invariant under all eight $D_4$ transformations. The group module constructs transformations from board coordinates rather than hand-written permutations. The model checks compare trainable parameter counts with the orbit structure, and equivariance checks verify that subgroup-equivariant models give identical predictions on transformed boards up to numerical tolerance. The deterministic rule oracle is included only as a sanity reference: it confirms that the task is rule-solvable and that imperfect variational accuracy is a property of the ansatz and optimization, not of ambiguous labels.

The most important remaining limitation is statistical. Five seeds are enough to identify the architecture signal and to draft the paper, but they are not enough for a final robustness claim. The final version should rerun the reproduction and oracle-inspired comparisons with more seeds, report bootstrap intervals or nonparametric paired tests, and include finite-shot simulations if the work is framed toward hardware. The current claim is intentionally narrower: within exact-state simulation, task-aligned equivariant gates substantially improve a Meyer-style symmetric VQLM on a controlled benchmark.

## Author Contributions

TODO.

## References


[1] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3:625–644, 2021.

[2] Adrián Pérez-Salinas, Alba Cervera-Lierta, Elies Gil-Fuster, and José I. Latorre. Data re-uploading for a universal quantum classifier. *Quantum*, 4:226, 2020.

[3] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. Effect of data encoding on the expressive power of variational quantum-machine-learning models. *Physical Review A*, 103:032430, 2021.

[4] Johannes Jakob Meyer, Marian Mularski, Elies Gil-Fuster, Antonio Anna Mele, Francesco Arzani, Alissa Wilms, and Jens Eisert. Exploiting symmetry in variational quantum machine learning. *PRX Quantum*, 4:010328, 2023.

[5] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.

[6] Ville Bergholm, Josh Izaac, Maria Schuld, Christian Gogolin, Shahnawaz Ahmed, Vishnu Ajith, M. Sohaib Alam, Guillermo Alonso-Linaje, B. AkashNarayanan, Ali Asadi, Juan Miguel Arrazola, Utkarsh Azad, Sam Banning, Carsten Blank, Thomas R. Bromley, Benjamin A. Cordier, Jack Ceroni, Alain Delgado, Olivia Di Matteo, Amintor Dusko, et al. Pennylane: Automatic differentiation of hybrid quantum-classical computations. *arXiv preprint arXiv:1811.04968*, 2018.

[7] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[8] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.